Group number:

	NTNU Norges teknisk-naturvitenskapelige universitet Fakultet for Ingeniørvitenskap Institutt for marin teknikk	**EXERCISE 1** RESPONSE SHEET
TMR4182 MARINE DYNAMICS		

Approval of the exercise is based on three components:

1. Pre-lab calculations: Columns with gray headers should be filled out prior to the lab. The student assistant will only check to see if a reasonable attempt has been made (and may inform you if something appears wildly incorrect).
2. Participation: All group members should attend and be actively engaged. To be judged by the student assistant.
3. Responses to remaining questions on the handout. These can be brief - the purpose is to reflect on what you have done. To be submitted via Canvas.

1 TM161 Simple and compound pendulums

1.1 Experiment 1 - simple pendulum

$T = 2\pi\sqrt{\frac{l}{g}}$ $\quad \frac{T^2}{4\pi^2} = \frac{l}{g}$ $\quad g = \frac{l \cdot 4\pi^2}{T^2}$

Table 1: Results table for simple pendulum (light sphere)

Length (l) [m]	**Number of** n **swings** (n)	**Time for** n **swings** [s]	**Measured period** (T) [s]	**Theoretical period** (T) [s]	**Estimated gravity** (g) [m/s^2]
0.600	10	15.56	1.556	1.55389	9.783
0.400	10	12.43	1.243	1.2687	10.22
0.200	10	8.97	0.892	0.89714	9.92

Table 2: Results table for simple pendulum (heavy sphere)

Length (l) [m]	**Number of** n **swings** (n)	**Time for** n **swings** [s]	**Measured period** (T) [s]	**Theoretical period** (T) [s]	**Estimated gravity** (g) [m/s^2]
0.600	10	15.38	1.53	1.55389	10.11
0.400	10	12.37	1.237	1.2687	10.32
0.200	10	8.82	0.882	0.89714	10.15

- Compare measured and theoretical periods for lightest and heaviest sphere. Try to explain possible reasons for any differences you observe.

Air resistance and it beeing more difficult to meson the short one due to its higher velocity .

- Comment on the estimated acceleration due to gravity compared to reference standard values (or latitude-specific values).

the long and light one is Weirdly the most accurate which is unexspected due to Air resistance Becing a larger Part of the Total Energy in that case.

1.2 Experiment 2: The compound pendulum

Table 3: Results table for compound pendulum

Length (l) [m]	**Height** h [m]	I_A [kg/m^2]	**Time for** n **oscillations** [s]	**Measured period** (T) [s]	**Theoretical period** [s]
0.600	0.282	0.024911	12.57	1.257	1.27
0.500	0.182	0.014703	11.94	1.198	1.2156
0.400	0.082	0.0089504	14.08	1.408	1.40864

- Document your calculation of the moment of inertia of the rod.

$$I_{rod} = \frac{m_{rod} L^2}{12} = \frac{0.220\,(0.636)^2}{12} = 0.007416\ \text{kgm}^2$$

$$I_A = I_{rod} + m_{rod} h^2$$

- Compare the theoretical results and experimental results and comment on how good the agreement is.

theoretical and mesured varies with $-1.0\% \to -0.05\%$, therfor within normal variation for Exsperiments

- What would you expect to happen if $h = 0$? Why?

When $h \to 0$ $\quad T = 2\pi\sqrt{\frac{I_A}{mgh}} \Rightarrow T \to \infty$

1.3 Experiment 3: Compound pendulum with adjustable mass

Table 4: Results table for compound pendulum with adjustable mass

Length (l_2) [m]	**Height** h [m]	**Time for** n **oscillations** [s]	**Measured period** (T) [s]	**Theoretical period** (T) [s]	I_{TOT}
0.550	0.479	14.35	1.435	1.453	0.181
0.500	0.444	13.78	1.378	1.392	0.154
0.450	0.4096	12.92	1.292	1.337	0.131
0.400	0.37494	12.35	1.235	1.274	0.109
0.350	0.3402	11.95	1.195	1.222	0.091

- Compare your results for the simple pendulum, the compound pendulum, and the compound pendulum with mass. For the simple pendulum, the period should become shorter as the length of the pendulum below the pivot point becomes shorter. What happens for the compound pendulum? Why? What about the compound pendulum with adjustable mass?

the compond pendulum gets it's restoring momentum gets small when the pivot gets close to the centre of mass => The Period increases

With adjustable mass moving the sphee upwards reduces $h \Rightarrow I_{TOT}$ in our mesured range inertia reduction Dominates => T decreased from 1.435 -> 1.195

Why?

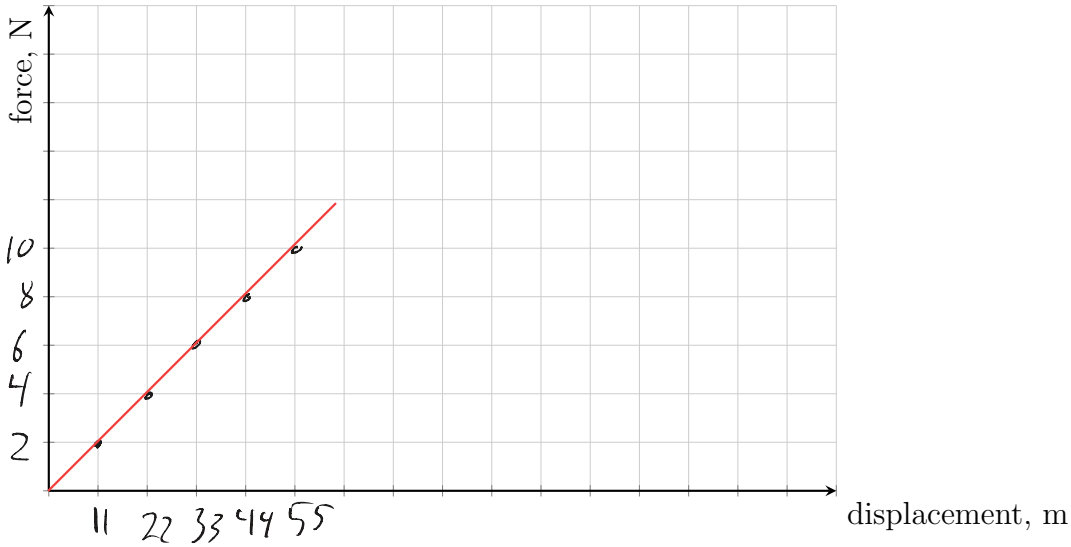
2 TM164 Free vibrations of a mass-spring system

2.1 Experiment 1: Static experiment

Table 5: Results table

Δ m [kg]	Δ F(Δmg) [N]	Scale value [~~m~~] [mm]	Displacement (Δ scale value) [~~m~~] [mm]
0	0	116	0
0.2	1.962	127	11
0.4	3.924	138	22
0.6	5.886	149	33
0.8	7.848	160	44
1.0	9.81	171	55

- Make a plot of force against displacement using your results. Do the plots confirm that the springs obey Hooke's law?



- From the plots of force against displacement, find the gradient and therefore the spring constant in N/m. Compare the result to the nominal value.

$K = 178.36\ N/m$

Nominal: 180 => Little bit less

2.2 Experiment 2: "Undamped" oscillation with varied mass

Table 6: Results table

m_{total} [kg]	T_0 (theory, $k = 180$ N/m)[s]	T_0 (theory, k_{meas}) [s]	T_0 (measured) [s]	$\frac{T_0\sqrt{k}}{2\pi}$	m_{total}^2	$\sqrt{m_{total}}$	$\frac{1}{m_{total}}$
0.605	0.364	0.366	0.372	0.791	0.366	0.3025	1.652
0.805	0.420	0.422	0.424	0.901	0.648	0.4025	1.242
1.005	0.4695	0.472	0.466	0.991	1.010	0.5025	0.995
1.205	0.5141	0.516	0.517	1.099	1.452	0.6025	0.830
1.405	0.5551	0.556	0.5664	1.204	1.974	0.7025	0.712
1.605	0.5933	0.596	0.600	1.275	2.576	0.8025	0.623

$2\pi \cdot \sqrt{\frac{m}{K}} = T_0$

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Table 7: Results table

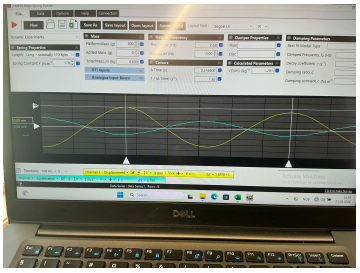
m_{total} [kg]	u_i [m]	u_{i+1} [m]	$\frac{u_i}{u_{i+1}}$	$\frac{1}{2\pi}\ln\frac{u_i}{u_{i+1}}$ **damping ratio(ζ)**
0.605	6.88	6.4	1.075	0.296
0.805	6.6	5.7	1.158	0.275
1.005	6.6	5.16	1.279	0.249
1.205	6.78	5.8	1.169	0.272
1.405	8.33	6.3	1.322	0.241
1.605	6.969	6.062	1.150	0.277
Average				0.268

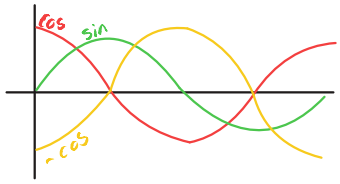
- Show a time series of measured displacement and acceleration. (Sketch based on the screen, or attach a screenshot or plot). Draw what you expect the velocity to look like. What is the phase difference between:
  - measured displacement and measured acceleration?
  - displacement and velocity?
  - velocity and acceleration?

For harmonic $x = A\cos(\omega t + \alpha)$ — x and $\ddot{x}$ 180°

$\Rightarrow \dot{x} = -\omega A \sin(\omega t + \alpha)$ — x and $\dot{x}$ 90°

$\ddot{x} = -\omega^2 A \cos(\omega t + \alpha)$ — $\dot{x}$ and $\ddot{x}$ 90°





- Why does the amplitude of the displacement decrease with time?

Due to the spring getting dampend, therfore going towards the equlibrium point

By Air reccistanse Friction => not id undamped

- Why is there a (small) difference between the measured natural period and the theoretical

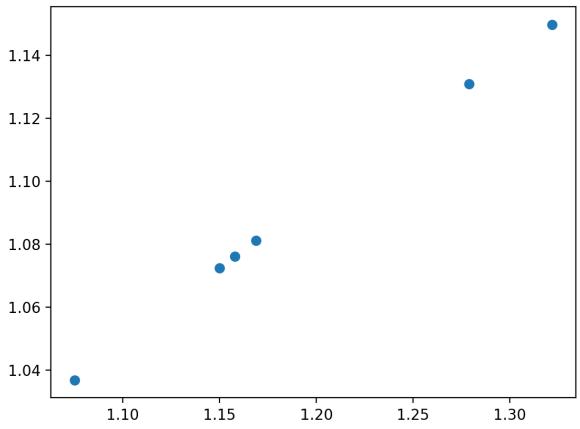
Uncertonities in the : Mesuring, Energy lost to heat, Air resitance

one?

- Make plots of $\frac{T_0\sqrt{k}}{2\pi}$ against each of the following:
  - m_{total}^2
  - $\sqrt{m_{total}}$
  - $\frac{1}{m_{total}}$

  Which curve appears linear?

$\sqrt{m_{TOT}}$ is linear



- Does the damping ratio (in air) depend on the mass of the system? Does the damping coefficient c depend on the mass of the system?

No, it seems to be constant

2.3 Experiment 3: Under-damped oscillations

Table 8: Results table for damper A. Use as many lines as needed!

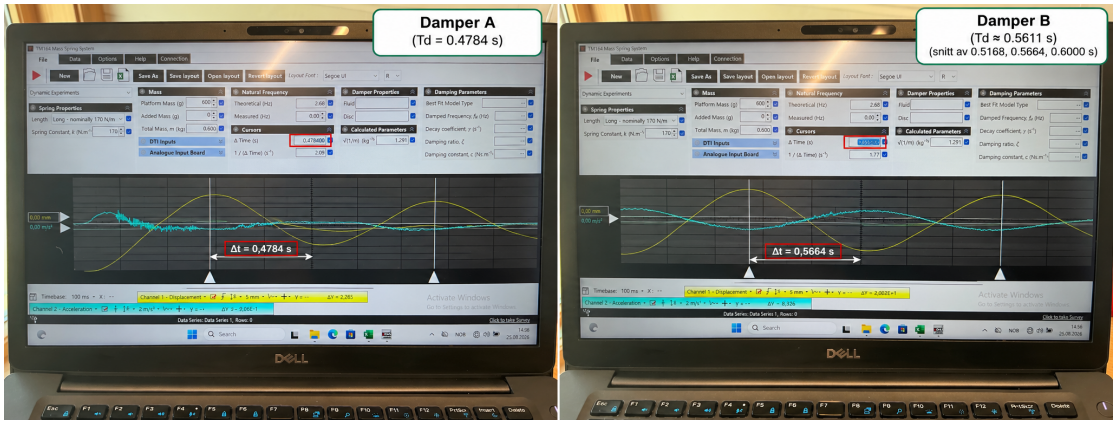
u_i [m]	u_{i+1} [m]	$\frac{u_i}{u_{i+1}}$	$\frac{1}{2\pi}\ln\frac{u_i}{u_{i+1}}$ damping ratio(ζ)
6.786	3.348	2.027	0.1124
7.056	3.528	2.001	0.1104
6.696	3.166	2.115	0.1192
Avg			0.1140

Table 9: Results table for damper B. Use as many lines as needed!

u_i [m]	u_{i+1} [m]	$\frac{u_i}{u_{i+1}}$	$\frac{1}{2\pi}\ln\frac{u_i}{u_{i+1}}$ damping ratio(ζ)
3.166	1.176	2.692	0.1576
2.714	0.683	3.974	0.2196
2.714	0.632	4.294	0.2319
~~1.72~~	0	$\ln\frac{x}{0}$ undef => excluded	
Avg			0.2030

Table 10: Results table. Note that T_0 "measured" should be found using the measured damped natural period and your estimate for the spring constant and damping ratio.

Damper	m (total) [kg]	T_0 (theory) [s]	T_d (measured) [s]	T_0 (measured) [s]	ζ (measured)
A	1.05	0.4821	0.4784	0.4753	0.1140
B	1.05	0.4821	0.5611	0.5494	0.2031



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- Explain the difference between the measured T_0 in Table 6 and Table 10

We get both Damping and added mass. the latter changes the effective oscillating Period, thefor the natural Period. The damping Also increases measured damping Period slightly than the corresponding Undamped

- How would you expect the natural period and damping ratio to change if we used oil instead of water in the damper? (You may assume that the oil would have a higher viscosity and lower density than water).

Oil is more Viscous than water, and would thefor give us an larger Damping Ratio, Ergo Larger Damping $\frac{T_d}{T_0}$ larger